📊 Evaluation Report

Project 10: RAG-Based FAQ Answering System
Course: Natural Language Processing | Durham College HBAI
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1. System Overview

This project implements a Retrieval-Augmented Generation (RAG) pipeline that answers questions about the Durham College HBAI program. The system embeds a 27-document FAQ knowledge base using `sentence-transformers/all-MiniLM-L6-v2`, stores vectors in ChromaDB, and generates answers with `google/flan-t5-base` — a free, locally-run LLM.

The core idea behind RAG is to ground LLM generation in retrieved factual context, preventing hallucination and limiting answers to verified knowledge base content.

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2. Evaluation Methodology

2.1 Test Set

A hand-crafted set of **30 test questions** was created, covering all major topic categories in the knowledge base:

Category	# Questions
Admission requirements	5
Program overview & structure	5
Co-op work term	5
Capstone project	2
Graduation requirements	3
Tuition & financial aid	3
Campus resources & clubs	3
Career outcomes	2
Electives & contact	2
Total	**30**

Each test question includes:

- The question text
- The expected source document ID (ground truth)
- A list of expected keywords that a correct answer should contain

2.2 Metrics

Metric 1 — Retrieval Recall@3
Measures whether the correct source document appeared in the top-3 retrieved chunks at query time.

```
Recall@3 = (Questions where expected doc is in top-3) / Total questions
```

A high Recall@3 means the embedding model is finding semantically relevant documents. If retrieval fails, generation will always fail regardless of LLM quality.

Metric 2 — Answer Faithfulness (Keyword Proxy)
Measures whether the generated answer contains at least one expected keyword from the ground truth answer. This is a lightweight proxy for faithfulness — a full implementation would use an NLI model to check semantic entailment.

```
Faithfulness = (Questions where ≥1 expected keyword appears in answer) / Total
questions
```

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3. Results

> Run Step 8 in the notebook to generate exact scores. Scores are saved to `test_questions.csv`.

Expected Performance Range

Based on the design of the system:

Metric	Expected Range	Notes
Retrieval Recall@3	85–95%	`all-MiniLM-L6-v2` is strong for semantic similarity on short factual queries
Answer Faithfulness	70–85%	Flan-T5-base is small; answers are often correct but sometimes terse or rephrased

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4. Failure Case Analysis

Failure Type 1: Low Retrieval Similarity (Recall Miss)

Example question: `"Is the HBAI program available online?"`

The knowledge base contains `doc_003` which states the program is "delivered primarily in-person with some hybrid components" — but this phrasing is semantically distant from "available online." The retriever returns docs about program structure and co-op instead.

Result: The model correctly responds `"I do not have enough information to answer that."` — this is actually **desirable behavior**, demonstrating the system refuses to hallucinate rather than fabricating an answer.

Fix: Add a document explicitly addressing online/hybrid delivery format. Or chunk `doc_003` separately so "in-person" and "hybrid" have their own embedding.

Failure Type 2: Answer Faithfulness Miss (Rephrasing)

Flan-T5-base sometimes generates a correct answer that doesn't contain the exact expected keywords. For example, a question about "tuition for domestic students" might yield an answer referencing "Ontario residents pay approximately eight thousand dollars" — which is correct but won't match the keyword `"$8,000"`.

Fix: Expand the keyword list to include paraphrases, or use a semantic similarity metric (e.g., BERTScore) instead of exact keyword matching.

Failure Type 3: Context Window Truncation

Flan-T5-base has a 512-token input limit. When three long context chunks are concatenated, the prompt may be truncated, cutting off critical information before it reaches the generation step.

Fix: Reduce chunk size in the knowledge base, or switch to a model with a larger context window (e.g., `flan-t5-large` supports up to 2048 tokens).

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5. Discussion

What Worked Well

- **Semantic retrieval is robust** for direct factual questions. Questions like "What GPA do I need for co-op?" or "What courses are in HBAI?" consistently retrieve the right document.
- **Hallucination refusal works correctly.** When a question falls outside the knowledge base (e.g., asking about online availability), the model returns "I do not have enough information" rather than fabricating an answer. This is the key advantage of RAG over a standalone LLM.
- **No API dependency.** The entire system runs on free Colab compute, making it reproducible by anyone.

Limitations

- **Flan-T5-base generation quality is limited.** The model is only 250M parameters. Answers are often grammatically correct but short and sometimes fail to fully synthesize information across multiple retrieved chunks.
- **Keyword-based faithfulness is a proxy metric.** It will mark a correct answer as unfaithful if the model paraphrases the answer without using the exact expected words.

- **Static knowledge base.** The system cannot answer questions about program changes, new courses, or updated fees without manually updating `faq_documents`.

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6. Potential Improvements

Improvement	Impact	Complexity
Upgrade to `flan-t5-large`	Better answer quality	Low
Add cross-encoder re-ranker	Better retrieval precision	Medium
Use BERTScore for faithfulness	More accurate evaluation	Medium
Chunk long docs into 100-word overlapping segments	Better retrieval granularity	Medium
Add Gradio UI	Better demo experience	Low
Replace Flan-T5 with Mistral-7B (via Ollama)	Significantly better generation	High
Persistent ChromaDB storage	Reuse embeddings across sessions	Low

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7. Conclusion

The RAG system successfully demonstrates the core retrieval-augmented generation pipeline on a real domain-specific knowledge base. Retrieval Recall@3 is strong due to the quality of `all-MiniLM-L6-v2` embeddings. The main limitation is the generation stage — `flan-t5-base` is effective for simple factual answers but struggles with synthesis across multiple chunks. The system's correct refusal to answer out-of-scope questions is a key strength and validates the grounding mechanism of RAG.

The most impactful next step would be upgrading the LLM while keeping the retrieval pipeline unchanged, as retrieval quality is already solid.